

Identify maximums, minimums, and where a function increases or decreases

Pre-Cal | CK-12: Maximums and Minimums

Key Idea

A **maximum** is a highest point; a **minimum** is a lowest point. A function **increases** where the graph rises left-to-right and **decreases** where it falls.

Key Terms

- **Extremum** — a max or min point
- **Interval** — the stretch of x -values where a behavior holds

Worked steps

Step 1: Find the turning point (vertex or peak/valley).

Step 2: Read increasing/decreasing on each side.

Example: $y = x^2$ has a **minimum** at $(0, 0)$; it decreases for $x < 0$ and increases for $x > 0$.

You Try

Does $y = -x^2$ have a max or a min, and where?